University of Arizona

Computer Science

CSC 380: Statistics and Data Science

Analysis of Tucson Crime and Income

By: Ashiqul Alam and Teddie Anest

Professor Cesim Erten

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Introduction

The goal of this project was to explore and model how crime can be affected by the certain demographics of neighborhoods. Crime is a concern that many citizens have when choosing where to move, attend school and raise their families, along with choosing how much precaution to have when navigating their daily life. This project focused on finding the relationships and correlations that exist between neighborhood income and crime rates by exploring two databases from the city of Tucson, Arizona: Arrest Data and Neighborhood Income (2019). Specifically, we looked at income markers such as the wealth index (a composite measure that estimates a household's relative economic status by analyzing their assets, housing and water access), median and per capita income, along with arrest markers such as number of arrests and crime type.

We aimed to find a relationship between income and crime to help determine which neighborhoods are prone to a high number of arrests or a certain type of crime. Additionally, we aimed to investigate whether predictive models could be created to estimate arrest and crime data based on neighborhood income data, which could serve as a valuable tool for decision-making for citizens and police departments. Citizens could use this information to make more informed choices about where to live or visit, and policymakers and law enforcement agencies could utilize these insights to prioritize resources for areas at greater risk. Ultimately, understanding this relationship could offer a more strategic approach to addressing crime and supporting community safety.

Related Works

There have been many studies over the years that have looked at overall income and crime trends in the world, along with specific countries such as the United States. After gathering numerous sources, there does not seem to be a definite answer among social scientists and statisticians about whether income has an effect on crime as a whole. It is a complex question to answer, as there may be numerous underlying factors at play. However, there are trends that have been found in the data. According to research professor Steven Levitt, an American economist at the University of Chicago, property crime specifically was shown to be concentrated in poor areas (property crime is defined as a category of crime that involves the theft or destruction of someone else's property), while violent crime (murder, assault, etc.) does not hold this pattern. On a larger scale, a 2002 study by World Bank concluded that "crime rates and inequality are positively correlated within countries and also between countries." It is important to note that there exist studies that state there is no relationship between income and crime, and deem other factors and demographics more significant.

Through our research, we looked at multiple income factors, looked at a more local scale (individual neighborhoods and specifically Tucson), and specific types of crime more closely, which seems to be lacking in most current research. When researching the current predictive models, there has been some work put into generating models that help predict crime rates. Most of this prediction relies on other socioeconomic factors, such as race and age, but we aim to offer a prediction on income that when used in conjunction with other research, could deem overall crime prediction more trustworthy (Aggarwal).

Methods and Results

The questions that this research aimed to answer are :

1. Does a higher income in a neighborhood result in more or less arrests?
2. Is there a relationship between certain types of crime in a neighborhood and income?
3. Can we predict the amount of arrests based on income?
4. Can we predict the most likely type of arrest when given income?

We first gathered our data from two databases, Tucson Police Arrests and Neighborhood Income. Then, "data cleaning" was performed on these two datasets before data exploration could begin, as there were invalid entries (for example, arrests that had no associated neighborhood and crimes that needed to be grouped together, such as car theft and shop theft.). Also, the exploration was narrowed to only certain features of the dataset, as seen below.

Tucson Police Arrests	**Neighborhood Income**
NHA_NAME: neighborhood name	NHA_NAME: neighborhood name
chrgdesc: description of arrest charge	WLTHINDX: wealth index
	TOTHH_CY: number of households
	PCI_CY: per capita income
	MEDHINC_CY: median household income

Then, by applying data science techniques and implementing Python code, we could begin analysis of the data, create visualization, and implement prediction models.

Exploratory Analysis and Visualizations

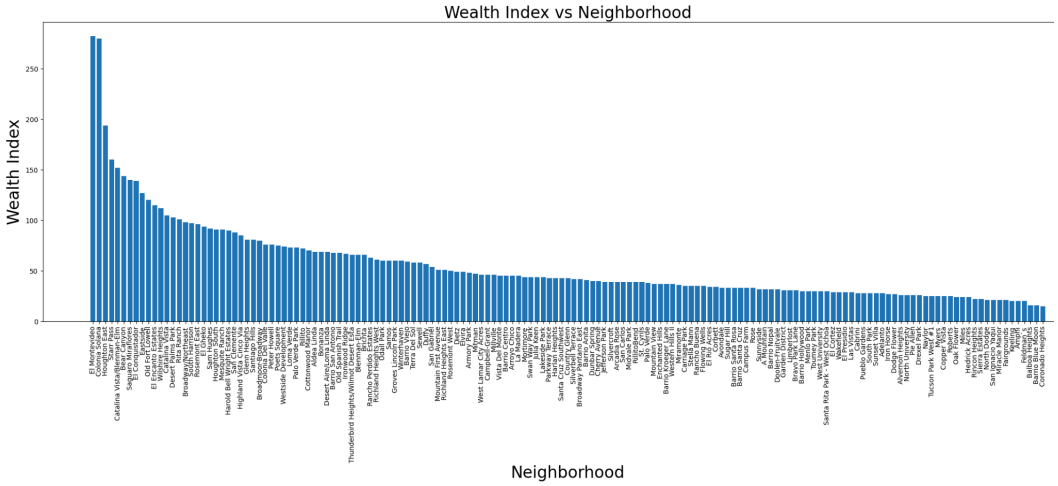


Figure 1

Though it contains lots of data, the first graph (Figure 1) was created to get an overall idea of the variety of wealth indices in the city of Tucson. The plot was created from the neighborhood income database, which contains 159 records. Each of the 159 neighborhoods was plotted on the x-axis. The wealth index ranges from 0 to 282 for Tucson neighborhoods, and is plotted on the y-axis (exact scale is unknown, as factors for calculating the value can vary). There seem to be two distinct upper outliers, El Montevideo and Colonia Solana, which have a wealth index of 282 and 280 respectively and the data has a big spread overall.

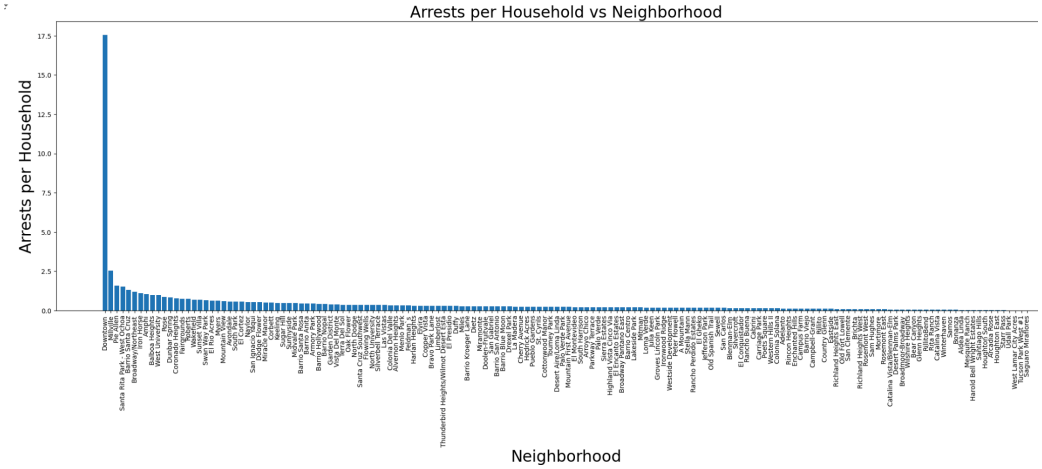


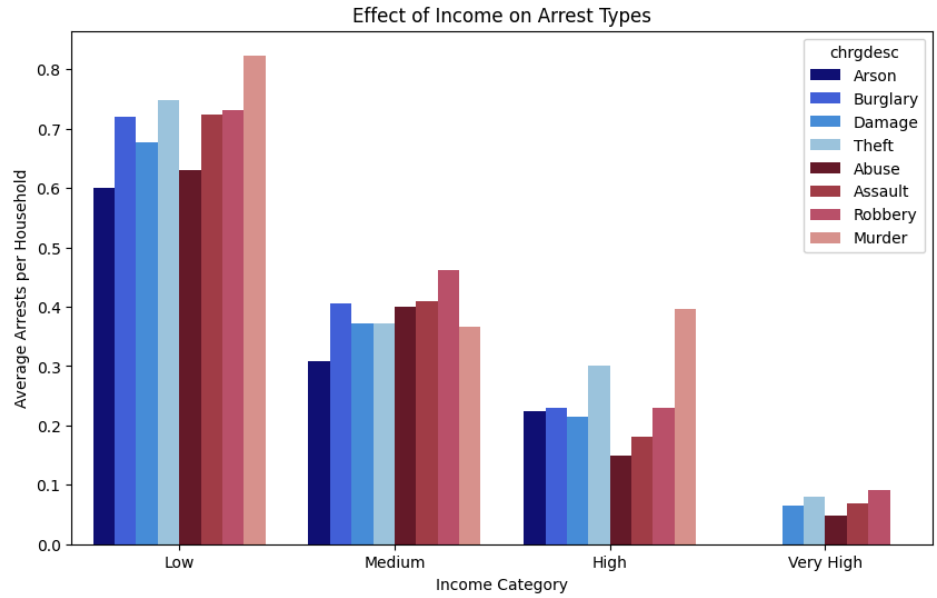
Figure 2

Figure 2, though it also has a lot of data, gives a good overview of the arrests in Tucson. The neighborhoods are plotted on the x-axis and the number of arrests per household (to account for varying population levels) is on the y-axis. There is a very prominent outlier, the Downtown neighborhood, which has an extremely large number of arrests per household (over 17). The rest of the neighborhoods begin at about 2.5 arrests per household, and continue down, with some neighborhoods recording 0. After analyzing the graph, the data showed that the downtown area was a huge single outlier in the data. After consideration, we considered possible explanations for this to be a low household number, but a large population (homeless, visitors, etc.), thus naturally increasing the possibility for arrests/ crime. In future plots, downtown is excluded from the data.

Figure 3

The graph in figure 3 is a heat map of the relationship between arrests per household and the wealth index, per capita income and median household income. The overall correlation coefficient found for arrests per household and the wealth index was -0.35, which shows that

there is some negative linear correlation between the variables: as wealth index increases, the number of arrests per household goes down. The correlation of per capita income was slightly more correlated at -0.41, illustrating that as per capita income goes up, the number of arrests per household decreases. Finally, median household income had the strongest correlation at -0.43, similarly showing that as median income increases, the number of arrests per household decreases.



Low: $0.00 - $29,000.00
Medium: $29,000.00 - $58,000.00
High: $58,000.00 - $87,000.00
Very High: $87,000.00 - $116,000.00

Figure 4

From research, it was clear that it may be possible that certain crimes could be more associated with certain incomes than others. So, we decided to look at median household income against certain types of crime. Eight crimes were picked and split into two categories, property (seen in the blue shades) and violent (seen in the red shades). Median household income was also split into four levels, which can be seen above. Figure 4 displays low median income produces the most arrests per household for each type of crime and a very high income produces the least

amount of arrests per household for each crime. When looking at specific crimes, it is shown that a property crime, such as burglary, goes down steadily as income increases. However, a violent crime like murder stays pretty high, even when income is classified as medium or high.

Prediction Models Analysis (Linear Regression)

After analyzing the data above, we focused on question three. Can we predict the arrests per household of a neighborhood if we know the income? The model that we chose to implement was Linear Regression. Before doing this, some additional "cleaning" had to be done. As illustrated from Figure 3, there is a negative correlation between all three measures of income, wealth index, per capita income, and median income, vs arrests per household. However, the three income variables are also highly correlated with each other (.74 to 1) so they were implemented as a single variable. To do this, the data was standardized (meaning the income features were scaled to have a mean 0 and standard deviation 1) and PCA was applied (to combine the features). After training was performed using this model, an equation for the line was produced, and the r2 score was reported (explains how well the regression line fits the data; 0 to 1).

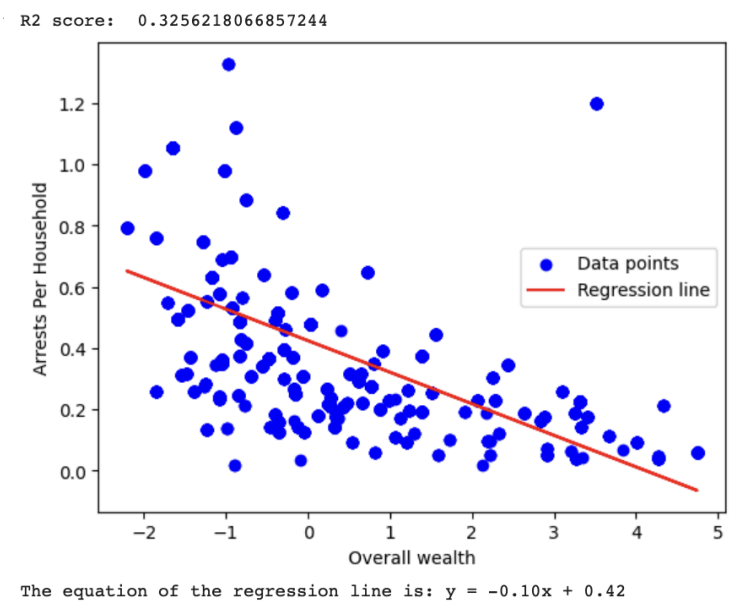


Figure 4

The negative slope of the line (-.10x) indicates that there is a negative correlation between the x and y variables, overall income and arrests per household, which is consistent with

the findings in the exploratory analysis. Specifically, if x (the overall income) increases by 10 units, the number of arrests per household would decrease by one. The r2 value of 0.32 displays that 32% of the variation in the y variable, arrests per household, can be explained by the x variable, the overall income. A polynomial regression model was also fit to the data, to see if an equation with degree two could better explain the data.

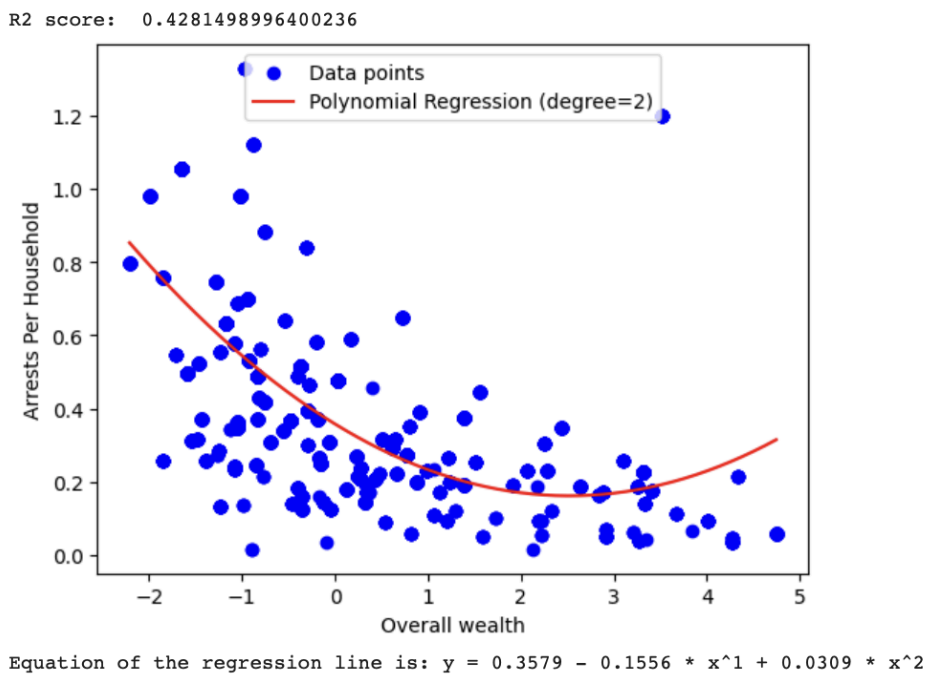


Figure 5

For this equation, the positive variable for the quadratic term indicates that as wealth increases significantly, the number of arrests per household would go up. The right hand side of the quadratic equation is not relevant to this data, however. The linear term, -0.1556, indicates that as wealth increases, the number of arrests per household decreases. This is consistent with the findings. Finally, the intercept indicates that if wealth is zero, there would be a small baseline number of arrests per household. The r2 score of this line was slightly higher than the linear model, 0.43, which illustrates that 43% of the variation in arrests can be explained by wealth. After the model was trained, the model was evaluated on test data.

```
Mean Squared Error on Test Set: 0.04516467569043462
R^2 Score on Test Set: 0.3522051271174287
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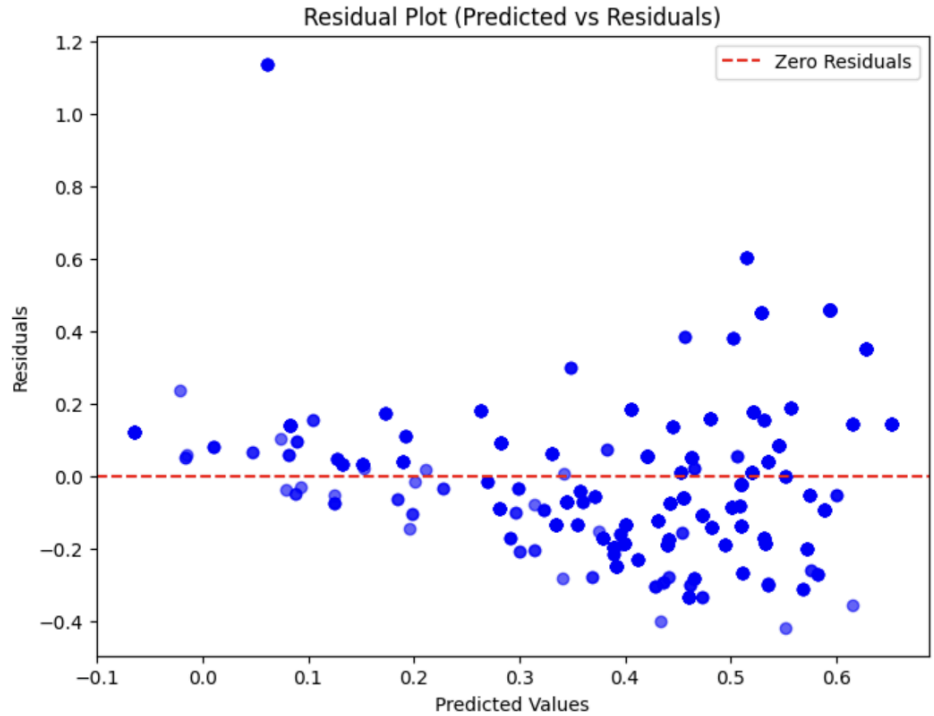
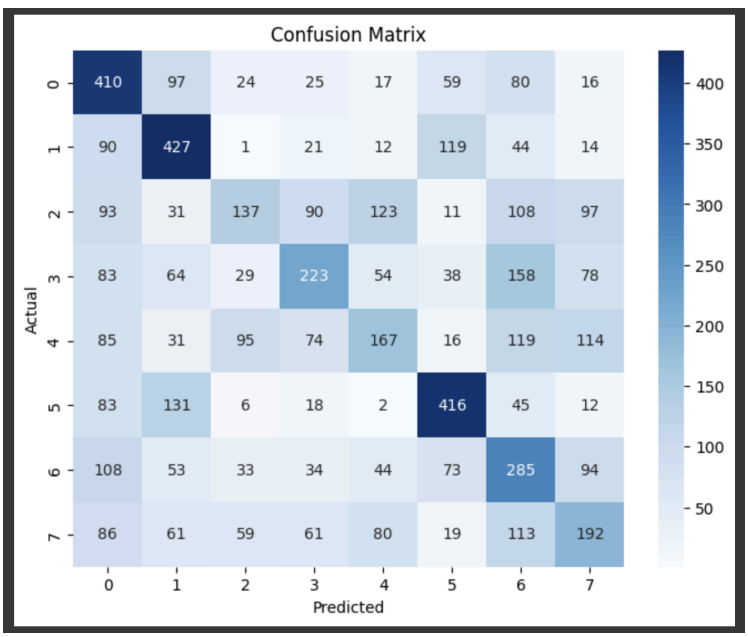


Figure 6

The residual plot in figure 6 helps visualise the differences between the actual values and what they were predicted to be. The r2 value for the test data is 0.35. This suggests that the model only accounts for 35% of variation in arrests per household. The model is limited in its ability: possible explanations are discussed in the conclusion below.

Prediction Models Analysis (Trees)

Finally, we focused on question four. Can we predict the most likely type of arrest when given income? The model that we chose to implement was a decision tree, which is used for classification. We first looked at the count of each of the eight crimes chosen (same as in Figure 4) and there appeared to be a class imbalance (some crimes only had 26 arrests while others had 3000). This was adjusted using Synthetic Minority Over-sampling Technique, instead of PCA. Instead of combining the income variables, like for Linear Regression, only median household income was used for this model. The decision tree was trained and the confusion matrix and result metrics were obtained.




```
Accuracy: 0.40
Classification Report:
              precision    recall  f1-score   support

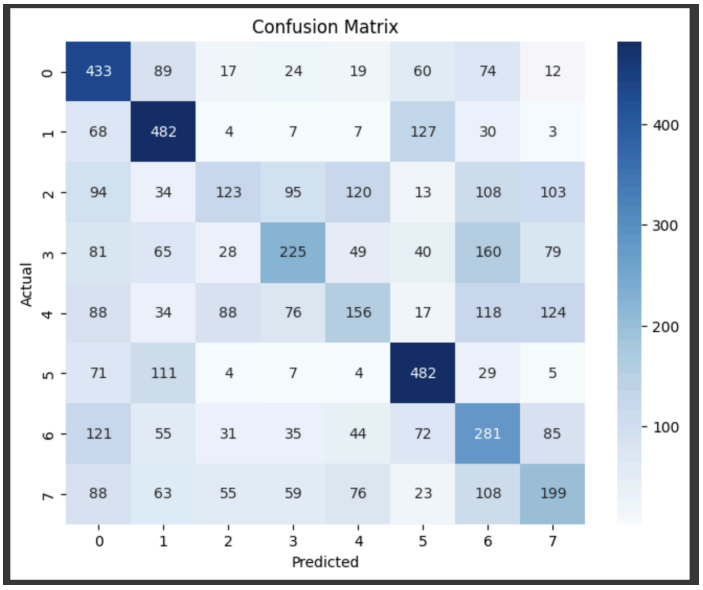
           0       0.39      0.56      0.46       728
           1       0.48      0.59      0.53       728
           2       0.36      0.20      0.26       690
           3       0.41      0.31      0.35       727
           4       0.33      0.24      0.28       701
           5       0.55      0.58      0.57       713
           6       0.30      0.39      0.34       724
           7       0.31      0.29      0.30       671

    accuracy                           0.40      5682
   macro avg       0.39      0.39      0.39      5682
weighted avg       0.39      0.40      0.39      5682
```

Figure 7
Abuse: 0, Arson: 1, Assault: 2, Burglary: 3, Damage: 4, Murder: 5, Robbery: 6, Theft: 7

Visually, the confusion matrix displays a good "diagonal," meaning when the model is tasked with predicting a crime when given median household income, it does predict that crime more than any other crime. For certain crimes, specifically abuse, arson, burglary and murder, the model does better. The results could be further evaluated when looking at metrics such as

accuracy, as well. The accuracy overall found was 40%, which means that out of all predictions, the model predicted 38% of them correctly and 60% incorrectly. In order to attempt to increase the average, the Random Forest classifier was also implemented (a random subset of the features is used at each split in the tree).




```
Accuracy: 0.4190425906370996
              precision    recall  f1-score   support

           0       0.41      0.59      0.49       728
           1       0.52      0.66      0.58       728
           2       0.35      0.18      0.24       690
           3       0.43      0.31      0.36       727
           4       0.33      0.22      0.27       701
           5       0.58      0.68      0.62       713
           6       0.31      0.39      0.34       724
           7       0.33      0.30      0.31       671

    accuracy                           0.42      5682
   macro avg       0.41      0.42      0.40      5682
weighted avg       0.41      0.42      0.40      5682
```

Figure 8
Abuse: 0, Arson: 1, Assault: 2, Burglary: 3, Damage: 4, Murder: 5, Robbery: 6, Theft: 7

This model achieved slightly higher accuracy than the previous model overall, at 42%. It also performed the best for certain crimes (also abuse, arson, burglary and murder) which is similar to the previous model.

Conclusion

After performing exploratory data analysis (EDA), we addressed two prediction tasks: Can we predict the number of arrests per household in a neighborhood based on income? and Can we predict the most likely type of arrest given a neighborhood's income? For the first question, we applied Linear Regression, with the best-fit polynomial equation being $y=0.3579-0.1556x+0.0309x^2$, yielding an R2 of 0.43. Though 43% is reasonable, there is still a margin for improvement or possible explanations for this result. There may have been standardization issues with the three income variables (wealth index, per capita income and median household income). In the future, it might be better to focus on one variable at a time and aggregate the results later. This outcome also suggests that income alone may not be the most significant predictor of arrests per household; other factors likely contribute. Based on existing research and our own findings, income alone is not enough; demographic variables such as age, race, and location may play a substantial role.

The second model aimed to predict the most likely crime type based on income. The Random Forest classifier provided the best result, however there are explanations for the low accuracy as well. Like mentioned previously, income might not be the only factor that explains crime. Given the range of crime categories considered, some crimes, like theft, may be less correlated with income. In the future, it might be beneficial to focus on different types of crimes or incorporate additional features.

In both cases, in order to increase the accuracy of these models for Tucson, they should be trained on larger datasets/ multiple years to get a better representation. Also, our current classification of crime types may have been overly broad; for example, all assault-related crimes were grouped together. By collaborating with Police Departments, the type of crimes in the database could be better understood and categorized from their insights. It is clear that when

using factors like income, it should not be the only factor relied upon for predictive policing. In order to give departments and policymakers the best models, many different features of neighborhoods need to be considered. The implications of not having a large sample with many demographic factors increases the chance for bias, which could cause unnecessary targeting of certain communities.

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